Neural Chunking: Comparative Study of RNN and Transformer Architectures

COMP534 Assessment 3

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Introduction

Chunking (shallow parsing) labels every token with a BIO [10, 12] tag identifying phrase boundaries and types. Using a corpus of 8,919 sentences with 17 BIO labels, we compare six neural architectures spanning the recurrent and self-attentive families, with structured (CRF) and softmax decoders. The implementation uses only `nn.Linear`/`nn.Parameter` for the Transformer and a hand-written linear-chain CRF.

Data Management

Experimental Protocol

We adopt a *sentence-level* random 70/15/15 train/val/test split with seed 42. Splitting at sentence granularity prevents leakage of recurrent context across folds; token-level splitting would assign correlated tokens to different sets. The validation set is used for model selection and early stopping; the test set is held out and accessed exactly once. To strengthen the generalisation estimate we additionally run *5-fold cross-validation* on the (train $\cup$ val) pool for the best architecture, keeping the test set sealed — the standard *outer holdout* + *inner CV* protocol.

Dataloader and Preprocessing

The corpus parser reads two-column word/tag rows, treating blank lines as sentence delimiters; we recover 8,919 sentences (211,243 tokens). The *vocabulary is built on training data only* with min-frequency = 2, yielding $|V| = 7{,}755$; tokens below threshold map to `<UNK>`, giving an 8.95% test-set OOV rate. We reserve `<PAD>` $= 0$ for masking. Tags map to integer ids; PAD positions use label -100 so PyTorch's `CrossEntropyLoss(ignore_index=-100)` excludes them from gradients. Each batch uses *dynamic padding* via a custom `collate_fn`: tokens are padded to the longest sequence in the batch, with explicit boolean masks (`mask = ids != PAD`) threaded through every encoder, including `pack_padded_sequence` for RNNs and key-padding masks in MHSA. The class distribution is severely skewed (`I-NP`/`B-NP` together $\approx$ 39% of tokens; `B-INTJ`/`B-CONJP` occur <20 times), motivating macro-F1 and Cohen's κ in §3.

Neural Networks

Architectures

All six models share a PAD-aware learned embedding

- **BiLSTM** [3] — 1-layer bidirectional LSTM with packed sequences; classifier $\mathbf{W} \in \mathbb{R}^{2H\times T}$. The simple RNN baseline.
- **DeepBiLSTM** [8] — 2 stacked BiLSTM layers with inter-layer dropout, encouraging hierarchical representations: lower layer captures local boundary cues, upper integrates phrase-level context.
- **BiLSTM+CRF** [4, 6] — BiLSTM emissions feeding a *manual* linear-chain CRF [5]; adds a $T\times T$ transition matrix plus start/end vectors; trained by maximising the path log-likelihood $\mathcal{L} = -(\, e(y) - \log Z(\mathbf{x})\,)$ where Z is the forward-algorithm partition function; decoded with Viterbi.
- **Transformer** [14] — pre-norm encoder $\mathbf{x} \leftarrow \mathbf{x} + \mathrm{Drop}(\mathrm{MHSA}(\mathrm{LN}(\mathbf{x})))$ chosen for early-training stability [15]; sinusoidal positional encoding; attention uses key-padding masks. Built from `nn.Linear`/`nn.Parameter` only.
- **Transformer+CRF** — same encoder, structured decoder; allows a decoder-only ablation against the plain Transformer.
- **TF(learned-PE)** — replaces sinusoidal with a trainable `nn.Embedding` over absolute positions; the PE-type ablation.

Training Components

- Loss: `CrossEntropyLoss` ignoring PAD for softmax models; CRF negative-log-likelihood for structured ones.
- Optimiser: AdamW ($\beta = (0.9, 0.999)$, $\lambda_{\mathrm{wd}} = 10^{-5}$); decoupled weight decay avoids the L_2 regularisation inconsistency of plain Adam [7].
- Scheduler: ReduceLROnPlateau on val macro-F1 (factor 0.5, patience 1).
- Gradient norm clipped at 5.0 (RNN) and 1.0 (Transformer) — Vaswani-recommended for from-scratch attention.
- Early stopping on val macro-F1 (patience 3).
- Dropout 0.3 (RNN) / 0.1 (Transformer).
- Epochs: 12 softmax / 10 CRF (CRF is $\sim 5\times$ slower per step due to $O(LT^2)$ forward).

Validation results and best model

Table 1 compares all six models. **DeepBiLSTM** is best on every metric (chunk-F1 0.892, macro-F1 0.736); 5-fold CV confirms stability (0.8919±0.0046). McNemar's test [2, 9] shows DeepBiLSTM significantly beats every other model ($p < 10^{-9}$). Figure 1 shows training dynamics: RNN models converge smoothly within 8–10 epochs while Transformers plateau lower.

Two findings emerge from the ablation (Fig. 2):

Table 1: Validation comparison (sorted by chunk-F1). BIO Valid: % sequences with no illegal BIO transitions; CRF decoders suppress illegal transitions via learned penalties (BiLSTM+CRF: 97.0%), softmax models have no such constraint (BiLSTM: 80.6%).

Model	Params	ChF1	MacF1	BIO
DeepBiLSTM	3.37M	**0.892**	**0.736**	88.0%
BiLSTM+CRF	1.79M	0.883	0.704	**97.0%**
BiLSTM	1.79M	0.874	0.703	80.6%
Trans.+CRF	1.26M	0.811	0.510	71.2%
TF(lrn-PE)	1.29M	0.717	0.446	12.0%
Transformer	1.26M	0.704	0.464	13.5%

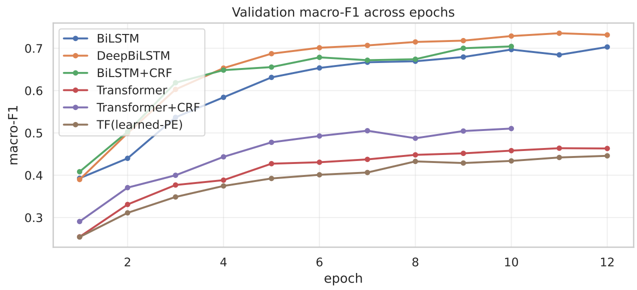

Figure 1: Validation macro-F1 per epoch (all six models).

- **Data Efficiency:** RNNs dominate Transformers by -0.170 chunk-F1, consistent with data-efficiency results [13] at $\sim$8.9k sentences.
- **Structural Priors:** Adding the CRF helps the Transformer ($+0.107$) far more than the BiLSTM ($+0.009$) — recurrence already imposes a soft sequential prior that CRF makes explicit.

Evaluate Models

Three orthogonal metrics

We report three metrics deliberately chosen to capture *distinct* aspects of the predictions:

- **Span-level chunk-F1** [11] — the CoNLL-2000 standard: a span is correct iff its (type, start, end) triple matches gold exactly; captures *chunk-correctness*;
- **Cohen's κ** [1] — chance-corrected agreement; captures *class-imbalance robustness* (a majority-class predictor scores high accuracy but $\kappa \approx 0$);
- **BIO sequence validity** — fraction of predicted sequences with no illegal transitions (e.g. `I-NP` after `O`); captures *structural correctness*.

None can be derived from another: a model can have high chunk-F1 yet low sequence validity, or high accuracy yet low κ.

3.2 Test-set results. Table 2 reports DeepBiLSTM on the held-out test set across the three metrics plus supporting diagnostics. The model achieves chunk-F1 $= 0.898$, $\kappa = 0.927$ (near-perfect chance-corrected agreement), and BIO validity $= 90.6\%$ (most sequences are structurally valid despite using a softmax head). Sentence exact-match is 44.8% — strict, since requiring all $\sim$24 tokens per sentence to be simultaneously correct under 94.1% per-token accuracy is exponentially demanding. Per-chunk-type scores show PP highest (0.970, closed-class prepositions are easily identified) and ADJP lowest (0.602, low support plus frequent ADJP$\leftrightarrow$ADVP confusion).

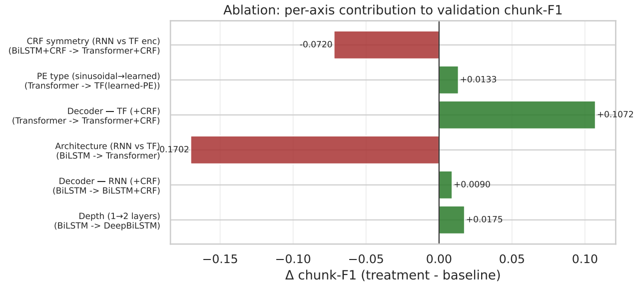

Figure 2: Ablation ΔChunkF1 for six single-axis comparisons.

Table 2: Test results — DeepBiLSTM.

Metric (aspect)	Score
Chunk-F1 (span correctness)	**0.8979**
Cohen's κ (imbalance-robust)	**0.9267**
BIO sequence validity (structural)	**90.58%**
Token accuracy	94.14%
Macro-F1	0.7278
Sentence exact-match	44.84%

3.3 Confusion matrix and error analysis. Figure 3 shows the chunk-type confusion matrix (the 17×17 BIO matrix is collapsed to 10 phrase families for readability). Diagonal mass exceeds 0.85 for the four major types (NP, VP, PP, SBAR). The two visible off-diagonal patterns are *ADVP$\leftrightarrow$ADJP* confusion (syntactically genuine ambiguity, e.g. *"well"* as ADVP/ADJP) and *ADJP$\rightarrow$O* false negatives. Decomposing the 1,855 token errors: 51.3% boundary errors (B$\leftrightarrow$I within the same family), 32.5% type errors (different families, same prefix), 8.9% O$\rightarrow$chunk and 7.4% chunk$\rightarrow$O — *segmentation, not classification, is the dominant failure mode*. The CRF transition matrix (Fig. 4) confirms valid BIO has been learned: B-X$\rightarrow$I-X transitions are the strongest preferred (scores $+0.85$ to $+1.09$) and O$\rightarrow$I-X transitions are penalised (-1.20 to -1.21), matching the BIO grammar.

Conclusion

DeepBiLSTM is the best model (chunk-F1 $= 0.898$, $\kappa = 0.927$, McNemar $p < 10^{-9}$). At 8.9k sentences, recurrence's inductive bias dominates global self-attention; the CRF closes part of this gap for Transformers. Hyperparameter search on DeepBiLSTM further raises CV chunk-F1 to 0.902, leaving subword/character encodings the most promising future direction for OOV-prone phrase types.

References

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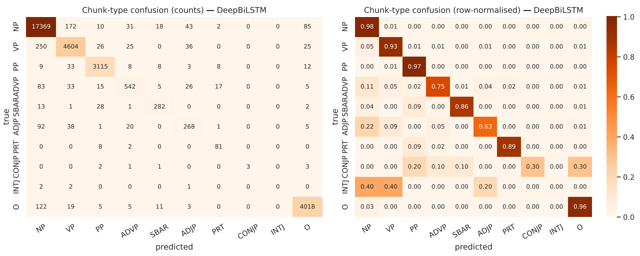

Figure 3: Chunk-type confusion matrix (row-normalised).

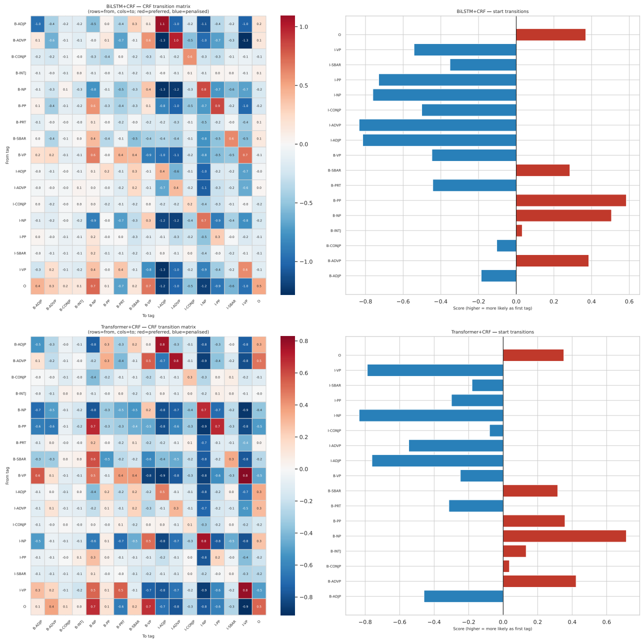

Figure 4: Learned CRF transition scores; red=preferred, blue=penalised.

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